

연세대학교
카피킬러캠퍼스 AI생성검사
요약 결과확인서

확 인
성 명
서 명

이메일	2025324176	AI작성률	24%
성명	자필로 기재하세요		
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비고			

분석 정보					
AI작성률	전체 영역	AI 의심 영역	높음	중간	낮음
24%	19	4	0	1	3

본 결과확인서는 2025324176 사용자가 카피킬러에서 수행한 문서 검사에 대한 분석 결과로 카피킬러 시스템이 생성한 자료입니다. 문서 작성과 평가에 대한 기준이 각 학교, 기관마다 다르므로 본 분석 결과와 최종 평가자의 평가는 다를 수 있습니다.

중간

Clinical Baseline Model For each disease outcome, we fit a baseline logistic regression model, termed clinic4, using the four clinical predictors introduced above: age, sex, height, and weight. Sex, a binary variable, was entered directly as a 0/1 indicator without scaling, while age, height, and weight were standardized to zero mean and unit variance using a scaler fit on the internal cohort and applied, without refitting, to the external cohort. This baseline was evaluated by internal AUC under 5-fold stratified cross-validation, reduced to as few as two folds when the minority outcome class contained fewer than five patients, and by external frozen AUC for HTN, DM, and CKD, serving as the reference against which the incremental value of AEC-128 was assessed.

① AI생성 특징

주관성의 지나친 배제

추상적, 일반적 내용 구성

무견해, 판단 회피적 성향

낮음

This was implemented as follows. The mean curve $\mu(z)$, for $z = 1, \dots, 128$, was computed across all curves in the internal cohort, and each patient's deviation curve was obtained by subtracting this mean from the raw curve, $d_i(z) = x_i(z) - \mu(z)$. The 128×128 covariance matrix C of the deviation curves was then eigen-decomposed via singular valuedecomposition (SVD) [7], using scikit-learn's PCA implementation for numerical stability. This decomposition takes the form $C \approx \Phi \Lambda \Phi^T$, where Φ is the $128 \times k$ matrix whose columns are the retained orthonormal eigenvectors (the eigenfunctions, denoted ϕ_k), Λ is the $k \times k$ diagonal matrix of the corresponding eigenvalues (denoted λ_k), ranked by the proportion of variance explained, and k is the number of retained components, determined below via an elbow-based criterion; truncating this product to the top k components yields the rank- k approximation of C used to assess reconstruction quality below. Each patient's deviation curve was projected onto each component to obtain the componentscore, $\text{score}_{i, k} = \sum_z d_i(z) \cdot \phi_k(z)$.

① AI생성 특징

지나친 요약 및 압축 서술

낮음

When fit to the internal cohort, PC1 alone explained 84.2% of the total variance; PC1-2 explained 94.7% cumulatively, and PC1-3 explained 97.3%. The number of retained components was determined using the Kneedle algorithm [8] applied to the explained-variance scree plot: the component index and explained-variance ratio were normalized to a 0-1 scale, and the elbow was defined as the point of maximal perpendicular distance from the line connecting the first and last components. This elbow coincided with PC3, fixing $k = 3$ as introduced above; the first three componentscores, PC1 through PC3, were accordingly retained as the representative AEC curvedescriptors. As an illustrative check of reconstruction accuracy at the individual level, one patient with large PC1-PC3 scores and reconstruction fit in the top quartile, with component values of PC1 +1,318, PC2 +1,175, and PC3 -247, was reconstructed from these three scores alone; the resulting R^2 between the reconstructed and original curve was 0.994, indicating that the 128-point curve can be closely approximated by three componentscores.

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낮음

For each disease, the resulting model combining clinic4 with the best-performing AEC featureset was compared against clinic4 by internal out-of-fold AUC and by external AUC from the internal-trained model applied once, without retraining, following the two-cohort clinic-model-versus-clinic-plus-curve-model design used in prior curve-based disease prediction studies [11]. The AUC difference between the two models was tested with the paired DeLong test [12], appropriate because both models were evaluated on the same patients. A classification threshold was chosen by maximizing Youden's J statistic [13] on internal out-of-fold predictions and then applied, fixed, to compute sensitivity, specificity, and accuracy in both cohorts. To assess whether the AEC contribution was consistent across CT manufacturers, AUC was additionally recomputed within each scannersubgroup containing at least 30 patients with both outcome classes represented, using the same fixed threshold. A two-sided p-value below 0.05 was considered statistically significant throughout. All analyses were performed in Python 3.13 using scikit-learn 1.7, pandas 2.3, NumPy 2.3, and SciPy 1.16.